

## EDUCATION

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>The Chinese University of Hong Kong (CUHK)</b>                                                                                                                                                                                        | Hong Kong SAR |
| <i>Doctor of Philosophy , Geophysics (Seismology)</i>                                                                                                                                                                                    | 2026 – 2030   |
| <b>Kharazmi University, Faculty of Earth Sciences</b>                                                                                                                                                                                    | Tehran, Iran  |
| <i>Bachelor of Science, Geology</i>                                                                                                                                                                                                      | 2021 – 2025   |
| <ul style="list-style-type: none"> <li>Chairperson, Geological Society of Kharazmi University (June 2024 – June 2025)</li> <li>Vice President of Media, Geological Society of Kharazmi University (November 2023 – June 2024)</li> </ul> |               |
| <b>University of Lille, International Academy Lille Hdf</b>                                                                                                                                                                              | Lille, France |
| <i>Water Resource Management Summer School (5 ECTS)</i>                                                                                                                                                                                  | Summer 2024   |
| <ul style="list-style-type: none"> <li>Presented a numerical groundwater modeling study on water abstraction impacts in the Weyib Watershed, Ethiopia. Under Supervision of the Prof. Arnaud Gauthier.</li> </ul>                        |               |

## TEACHING EXPERIENCE

|                                                                                                                                                       |                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CUHK</b>                                                                                                                                           | Hong Kong SAR                |
| <i>Hydrogeology Teaching Assistant</i>                                                                                                                | August 2026 – Present        |
| <ul style="list-style-type: none"> <li>(EESC3320), under supervision of Prof. Yen Joe Tan</li> </ul>                                                  |                              |
| <b>Space Research and Technology Institute (SRTI)</b>                                                                                                 | Primorsko, Bulgaria          |
| <i>Lecturer</i>                                                                                                                                       | August 2026 – September 2026 |
| <ul style="list-style-type: none"> <li>EPEC ML WG Summer School 2026, under supervision of Prof. Lachezar Filchev</li> </ul>                          |                              |
| <b>Iranian Young Scholars Club (IrYSC)</b>                                                                                                            | Tehran, Iran                 |
| <i>Earth Science Olympiad Teacher</i>                                                                                                                 | April 2021 – August 2026     |
| <b>Kharazmi University</b>                                                                                                                            | Tehran, Iran                 |
| <i>Programming Teaching Assistant</i>                                                                                                                 | April 2025 – June 2025       |
| <ul style="list-style-type: none"> <li>Programming Course, under supervision of Prof. Morteza Asemani</li> </ul>                                      |                              |
| <b>Kharazmi University</b>                                                                                                                            | Tehran, Iran                 |
| <i>Geostatistics Teaching Assistant</i>                                                                                                               | February 2025 – June 2025    |
| <ul style="list-style-type: none"> <li>Geostatistics Course, under supervision of Prof. Ehsan Pegah</li> </ul>                                        |                              |
| <b>Young Scholars Club</b>                                                                                                                            | Tehran, Iran                 |
| <i>Instructor</i>                                                                                                                                     | July 2024                    |
| <ul style="list-style-type: none"> <li>Taught "Applied Data Science in Geology" for the Iran National IESO<sup>1</sup> 2024 Team</li> </ul>           |                              |
| <b>Kharazmi University</b>                                                                                                                            | Tehran, Iran                 |
| <i>Instructor</i>                                                                                                                                     | April 2024 – June 2024       |
| <ul style="list-style-type: none"> <li>Organized and instructed the Advanced Python and Machine Learning Workshop</li> </ul>                          |                              |
| <b>NODET<sup>2</sup></b>                                                                                                                              | Tehran, Iran                 |
| <i>Teacher</i>                                                                                                                                        | May 2021 – September 2021    |
| <ul style="list-style-type: none"> <li>Taught ITEO<sup>3</sup> Marketing at Allameh Helli (Branches 5 &amp; 8) and Farzanegan High Schools</li> </ul> |                              |

## RESEARCH EXPERIENCE

|                                                                                                                                                        |                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>FEZtool</b>                                                                                                                                         | Tehran, Iran                |
| <i>Co-Founder &amp; Research and Development Specialist</i>                                                                                            | August 2023 – Present       |
| <ul style="list-style-type: none"> <li>Led Earth science research using advanced algorithms and geospatial analysis.</li> </ul>                        |                             |
| <b>Center for Complex Networks, SBU Partner</b>                                                                                                        | Tehran, Iran                |
| <i>Research Assistant</i>                                                                                                                              | November 2025 – June 2026   |
| <ul style="list-style-type: none"> <li>Worked on fashion trend forecasting project under supervision of Prof. Reza jafari.</li> </ul>                  |                             |
| <b>International Astronomical Search Collaboration, NASA Partner</b>                                                                                   | Remote                      |
| <i>Asteroid Search Team Member</i>                                                                                                                     | July 2025 – September 2025  |
| <ul style="list-style-type: none"> <li>Analyzed Pan-STARRS data, contributing to 3 preliminary asteroid discoveries.</li> </ul>                        |                             |
| <b>Kharazmi University</b>                                                                                                                             | Tehran, Iran                |
| <i>Research Assistant</i>                                                                                                                              | October 2024 – January 2025 |
| <ul style="list-style-type: none"> <li>Performed MASW<sup>4</sup> to extract dispersion images under supervision of Prof. Mehdi Talkhablou.</li> </ul> |                             |

<sup>1</sup>International Earth Science Olympiad

<sup>2</sup>National Organization for Development of Exceptional Talents

<sup>3</sup>Iranian Thinking & Entrepreneurship Olympiad

## COMMUNITY SERVICE

**Zamin Pouya Scientific Journal Vol. 31**

*Associate Editor & Reviewer*

**The 4<sup>th</sup> National Festival of Young Iranian Geologists**

*Education Committee Member*

**The 3<sup>rd</sup> National Festival of Young Iranian Geologists**

*Graphics Committee Member*

**Imam Khomeini International University**

*Editorial Board Member*

Tehran, Iran

*August 2025 – June 2026*

Tehran, Iran

*April 2025 – May 2025*

Tehran, Iran

*May 2024 – June 2024*

Qazvin, Iran

*September 2023 – June 2024*

## SELECTED PUBLICATIONS

### Journal Articles

1. Talkhablou, M., **Farmahini Farahani, M.**, & Mansouri, S. S. (2025). A Python-Based Framework for Land Cover Classification in Engineering Geology: A Comparative Assessment of SVM, K-Means, and Spectral Indices. *Kharazmi Journal of Earth Sciences*.
2. **Farmahini Farahani, M.** (2024). Designing the green belt in the southwest of Zahedan. *Ovan Scientific Journal*
3. Saravani, M. J., Kashef, S., **Farmahini Farahani, M.**, Kashefi, M., & Zohreh, M. (2023). Investigating the accuracy of hybrid models with wavelet transform in the forecast of watershed runoff. *Journal of Water Management Modeling (JWMM)*

### Conference Papers

1. **Farmahini Farahani, M.** (2025, August). Machine Learning-Enhanced Seismic Event Detection and Noise Reduction for Planetary Seismology: Insights from Mars and Lunar Data. In *Seismology Student Workshop (SSW 2025)*, GatherTown (Virtual)
2. Rezaei, K., & **Farmahini Farahani, M.** (2024, March). Visualization of magnetometric data with Python (case study: Segregation sedimentary units of Torud Playa). In *The 1<sup>st</sup> Conference on Applied Geophysics in Mines*, Tehran, Iran
3. Rezaei, K., & **Farmahini Farahani, M.** (2024, February). Comparing the performance of the k-means clustering method and spectral indices in characterizing land use and land cover using remote sensing with Python (case study: Landsat 8 multi-spectral images). In *2<sup>nd</sup> International Conference for Iranian Geography and Earth Sciences Students*, Tehran, Iran
4. **Farmahini Farahani, M.**, Rezaei, K., Karamollahi, S., & Moghadam, F. J. (2024, January). Application of PCA algorithm in remote sensing using Python programming language (case study: Landsat 9 multispectral images). In *8<sup>th</sup> Symposium of Sedimentological Society of Iran*, Hormozgan, Iran
5. Rezaei, K., & **Farmahini Farahani, M.** (2023, September). The importance of using Python programming language in Earth science. In *26<sup>th</sup> Conference of Geological Society of Iran*, Urmia, Iran

### Books

1. **Farmahini Farahani, M.**, Najibifar, Y., & Akbarzadeh Afshari, A. (2025). *Earthquake Prediction Using Geophysical Data Analysis and Artificial Intelligence*. **Heritage Branch, Library and Archives Canada**
2. Rezaei, K., **Farmahini Farahani, M.**, Karimzadehasl, E. (Trans.). (2024). *Introduction to Python in Earth science data analysis: From descriptive statistics to machine learning*. **Tasir Publication**. (Original work published by Maurizio Petrelli)

## PROJECTS

**PlanetaryGeophysics** | [github.com/aradfarahani/PlanetaryGeophysics](https://github.com/aradfarahani/PlanetaryGeophysics)

*2025 – Present*

- Built an open-source Python platform for analyzing and visualizing planetary geophysical data.

**FEZrs** (Open Source) | [pypi.org/project/fezrs](https://pypi.org/project/fezrs)

*2023 – Present*

- Advanced Python library for remote sensing, developed by FEZtool with over 100,000 downloads

**Seismicity Analysis in Iran (2010–2025)** | [iris.edu/hq/workshops/2025/06](https://iris.edu/hq/workshops/2025/06)

*2025*

- performed Iran seismicity analysis for the *Seismology Skill-Building Workshop 2025* final project.

<sup>4</sup>Multichannel Analysis of Surface Waves

- Interactive 3D visualization tool for subsurface resistivity using geoelectrics survey data

## INVITED TALKS

1. **Walrus: A Cross-Domain Foundation Model for Continuum Dynamics:** *Applied ML Lab, Center for Complex Networks, Department of Physics, Shahid Beheshti University*; 13 December 2025
2. **Applied Artificial Intelligence in Earth Science:** *Faculty of Geosciences, Shahrood University of Technology & Faculty of Earth Sciences, Kharazmi University*; 11 November 2025
3. **Applications of Artificial Intelligence in Education and Earth Science Analysis:** *Educational Technology Expert Group, Karaj and Fardis County Department of Education, District 2*; 30 December 2024

## LICENSES & CERTIFICATIONS

- Applied Geotechnical Seismology**, Kharazmi University 2026
- Applications of Non-Destructive Geophysical Methods in Engineering Geology Studies**, Kharazmi University
- Seismology Skill Building Workshop 2025**, EarthScope Consortium 2025
- Investigating practical issues in the exploration of Sulphide Deposits using the IP-RS method**, National Iranian Geophysical Society
- Remote Pilot Certification (Multicopter)**, Civil Aviation Authority of Islamic Republic of Iran 2023
- Unmanned Aerial Vehicle Surveying**, Maan Academy
- GIS, Mapping, and Spatial Analysis Specialization**, University of Toronto (Offered by Coursera)
- CS50 Certificate**, CS50 (Harvard University) 2021

## HONORS & AWARDS

- CUHK's International Student PhD Fellowship - Awardee**, CUHK 2026
- 9<sup>th</sup> National Young Scientists Festival - Merit Award Winner**, *Geology Category* 2025
- Hydro 2024 Research-Oriented Summer School - Full Scholarship**, University of Lille 2024
- Research Visit Grant (2500 DKK) - Awardee**, University of Copenhagen 2022
- DrCT 2021 Global Computational Competition - Gold Medalist** 2021
- 1<sup>st</sup> national Thinking & Entrepreneurship Olympiad (ITEO) - Silver Medalist** 2020
- WMI 2015 World Mathematical Championships Selection - Silver Medalist** 2015

## SKILLS & INTERESTS

**Programming Languages:** Python (ObsPy, pyGIMLi, Astropy, OpenCV, TensorFlow, PyGMT, Xarray, Scikit-learn, kepler.gl, Leafmap, Folium, Rasterio, GeoPandas), Julia, Bash (Shell scripting), SQL (MySQL), GMT, GDAL; Familiar with R

**Markup Languages:** Markdown, L<sup>A</sup>T<sub>E</sub>X

**Softwares:** Astrometrica, ENVI, Seismic Analysis Code(SAC), SeisAn, QGIS, SeisImager, Blender, ilastik, Google Earth Pro, Refrapy, SAS.Planet, draw.io, Groundwater Vistas

**Field Techniques:** Drone Surveying, Geoelectrical Methods (Induced Polarization–Resistivity Sounding, IP-RS), Seismic Refraction Data Acquisition

**Languages:** Farsi (Persian): *Native*; English: *Fluent (C1)*; German: *Elementary (A2)*; Mandarin: *Beginner (Duolingo Score: 10)*

**Research Interests:** Signal/Image Processing, Geophysics (Seismology), Remote Sensing, Machine Learning, Open-Source Software Development

## PROFESSIONAL MEMBERSHIPS

- |                                             |                                |
|---------------------------------------------|--------------------------------|
| <b>National Iranian Geophysical Society</b> | Student member, 2025 – Present |
| <b>Space Generation Advisory Council</b>    | Member, 2025 – Present         |
| <b>Google Maps Platform Innovators</b>      | Member, 2025 – Present         |
| <b>Geological Society of Iran</b>           | Student member, 2021 – 2022    |